AI Programming: Assignment4Part1

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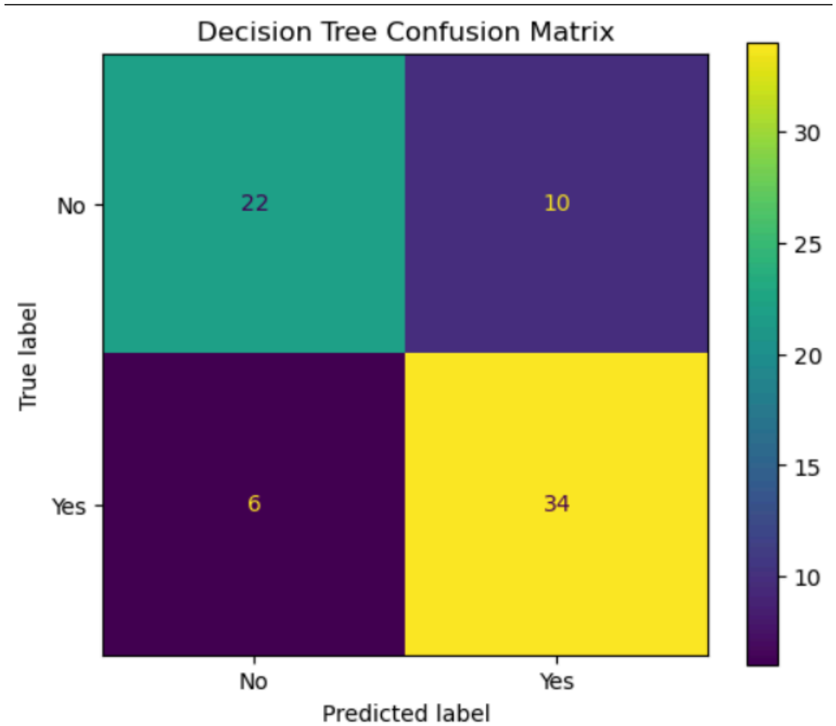
Results and Discussion

A Decision Tree classification model was developed to predict **Supplier_Delay_Risk** for ProcurePro Office. The objective of the model was to identify purchase orders that are likely to experience supplier delays, allowing the business to take proactive measures before delivery deadlines are missed.

Confusion Matrix Analysis



The confusion matrix provides a detailed view of the model's predictions. Out of 72 purchase orders in the test set, the model correctly classified 56 orders and incorrectly classified 16 orders.

Specifically, the model correctly identified 22 low-risk purchase orders (True Negatives) and 34 high-risk purchase orders (True Positives). However, it incorrectly classified 10 low-risk orders as high-risk (False Positives) and failed to identify 6 high-risk orders, classifying them as low-risk instead (False Negatives).

The confusion matrix shows that the model is generally effective at identifying delayed orders. This is particularly important because missing a genuinely high-risk purchase order could result in unexpected supply chain disruptions and delivery delays. The relatively low number of false negatives indicates that the model is able to capture most high-risk cases.

Classification Performance

The Decision Tree achieved an overall accuracy of **77.78%**, meaning that nearly four out of every five purchase orders were classified correctly.

The model obtained a **precision score of 77.27%**, indicating that when the model predicts a purchase order as high-risk, it is correct approximately three-quarters of the time. This helps reduce unnecessary interventions on low-risk orders.

The **recall score of 85.00%** is the strongest performance metric observed. Recall measures the model's ability to identify actual high-risk purchase orders. A recall of 85% means that the model successfully identified the majority of delayed orders, which is highly valuable from a business perspective because it minimizes the likelihood of overlooking potential supplier delays.

The model achieved an **F1-score of 80.95%**, demonstrating a good balance between precision and recall. This indicates that the model performs consistently across both identifying high-risk orders and avoiding excessive false alarms.

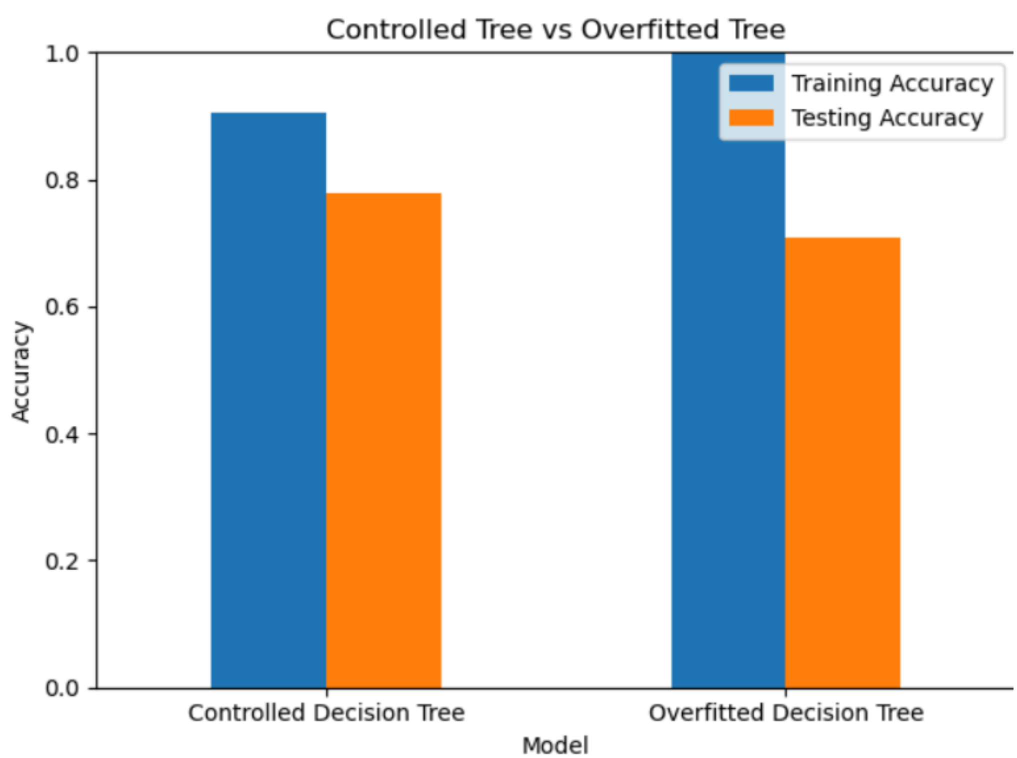
Class-Level Performance

The classification report provides additional insight into performance for each class.

For the "No" class (low delay risk), the model achieved a precision of 0.79, recall of 0.69, and F1-score of 0.73. This indicates that while the model performs reasonably well in identifying low-risk orders, some low-risk purchase orders are incorrectly flagged as high-risk.

For the "Yes" class (high delay risk), the model achieved a precision of 0.77, recall of 0.85, and F1-score of 0.81. The higher recall demonstrates that the model is particularly effective at detecting orders that are likely to be delayed.

Overfitting Analysis



To evaluate model generalization, the performance of a controlled Decision Tree was compared with an unrestricted Decision Tree.

The controlled Decision Tree achieved a training accuracy of **90.63%** and a testing accuracy of **77.78%**. While some performance drop is expected when moving from training data to unseen test data, the difference remains within an acceptable range.

In contrast, the overfitted Decision Tree achieved **100% training accuracy** but only **70.83% testing accuracy**. This substantial decrease in testing performance indicates that the model memorized patterns in the training data rather than learning generalizable relationships.

These results suggest that limiting the tree depth was effective in reducing overfitting and improving the model's ability to perform on unseen purchase orders.

Business Implications

The results demonstrate that the Decision Tree model can serve as a valuable decision-support tool for ProcurePro Office. By correctly identifying the majority of high-risk purchase orders, the model enables managers to focus attention on suppliers and orders that may require proactive intervention.

The high recall score is particularly beneficial because it reduces the risk of missing delayed orders, helping the organization improve supply chain reliability, reduce operational disruptions, and enhance customer satisfaction. Overall, the model provides meaningful predictive capability and can support more informed procurement and logistics decisions.